

## EXPERIENCE

- Rural Studies Institute, GCSU (Sponsored by Georgia Tech PIN Program)** Milledgeville, GA  
*Full-Stack Developer & Data Pipeline Intern* May 2025 – Aug 2025
- Developed custom **Python** scrapers as part of a **3-person team**, aggregating socio-economic datasets from US Census (ACS) and USDA APIs, automating ingestion of **10k+** data points and cutting manual data entry by **40%**.
  - Built automated data pipelines converting raw spreadsheets to **GeoJSON**, reducing integration time by **30%**.
  - Designed and deployed interactive map-based web platform using JavaScript, Leaflet.js, and WordPress; launched to **500+** monthly users across research and policy teams. Presented at the *Emerging Innovators Celebration*.
  - Project Link:** gcsu.edu/ruralstudies/understanding-black-belt-region

## TECHNICAL SKILLS

- Core (Proficient):** JavaScript/TypeScript, Python, Go, Node.js, Express.js, React.js, MongoDB, Git, REST APIs
- Cloud & Infrastructure:** AWS (**EC2, S3, Lambda, RDS, IAM**), Docker, Linux, CI/CD (GitHub Actions)
- Backend & APIs:** FastAPI, WebSockets/Socket.io, JWT, Microservices, OAuth concepts, JSON
- Databases & Data:** PostgreSQL, Prisma ORM, schema design, indexing
- Testing & Quality:** Jest, Vitest, Unit & Integration Testing, TDD
- Frontend:** Next.js, Tailwind CSS, Recoil, Zustand, Vite, HTML/CSS
- Tools, Practices & AI:** Postman, Zod, Bcrypt, Claude Code, Agile/Scrum, Code Reviews, Technical Documentation

## PROJECTS

- Real-Time Collaborative Task Board** | *React, Node.js, Express, Socket.io, MongoDB, JWT* [GitHub](#) | [Demo](#)
- Built real-time collaboration system using **WebSockets**, enabling instant task updates across **50+** concurrent users with **<100ms** latency.
  - Designed **RESTful API** with **JWT** authentication, **Zod** validation, **Bcrypt** password hashing, and role-based access control.
  - Improved data consistency by implementing a conflict resolution strategy for concurrent edits, reducing state synchronization errors by **15%** during peak activity.
  - Deployed on **Vercel** (frontend) and **Render** (backend) with automated **CI/CD via GitHub Actions** and health monitoring.
- TweenGen** | *React.js, Fabric.js, Recoil, Vite* [GitHub](#) | [Demo](#)
- Architected a lightweight animation engine with optimized easing algorithms, reducing the final bundle size by **15%** compared to standard external animation libraries.
  - Reduced canvas re-render cycles by **40%** through memoization and selective state updates in Recoil, maintaining a consistent **60 FPS** during complex multi-object animations.
  - Built drag-and-drop timeline UI with GSAP-powered live preview and code export system generating production-ready **HTML/CSS/JavaScript**; used **Claude Code** as a pair-programming reviewer to maintain a project logic graph and stress-test edge cases. Hosted on **Vercel**.

## EDUCATION

- The University of Georgia** Athens, GA  
*Master of Science in Computer Science* Aug 2024 – May 2026
- Coursework: Software Engineering, Database Management, Algorithms, Computer Networks, Data Science, Advanced Representation Learning
- Sathyabama Institute of Science and Technology** Chennai, India  
*Bachelor of Engineering in Computer Science* Aug 2020 – May 2024

## CERTIFICATIONS

- AWS Certified Developer – Associate ([link](#))
- AWS Certified Cloud Practitioner ([link](#))